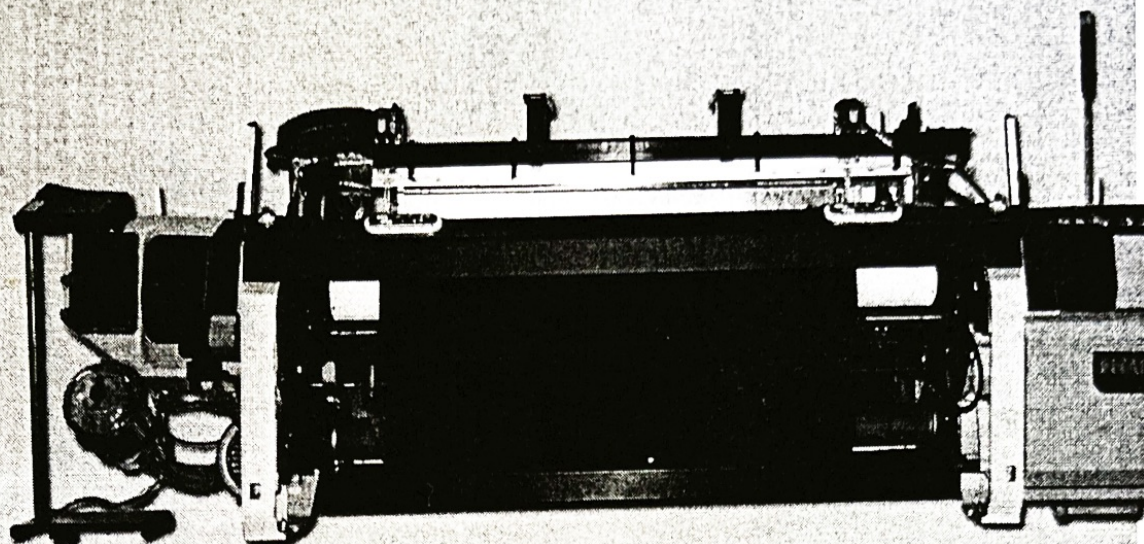


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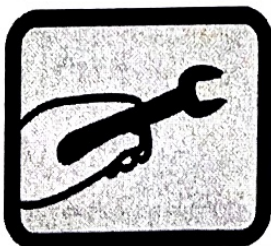
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Wiring and electronic control

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Content:

Pictographs, abbreviations and safety devices.....	9
• Pictographs used	9
• Abbreviations	9
• Safety Devices	9
General	11
Connection conditions	13
• Power supply cable specifications.....	13
• Supply cable protections	15
Control system and machine drive	17
• Principle	17
• Security	17
• Power supply - motor switches.....	18
• Control system of the main motor control box	18
• Main motor	18
• Brake.....	18
• Cooling system.....	19
• Pickfinding.....	19
• Weaving (driving the machine).....	20
• PXWP and PXPFP	20
• Let-off	21
• Take-up motion	24
Overview of the weaving machine.....	25
• Overview of weaving machine 1	25
• Overview of weaving machine 2.....	25
• Overview of weaving machine 3.....	26
• Overview of weaving machine 4.....	26
Main control box (CB) > Parts on the outside of the main control box	27
• Overview of the outside of the main control box	27
• Side 1	27
• Side 2	28
• Side 3	28
• Side 4	29
Main control box (CB) > High-voltage section	30
• General	30
• Components.....	31
• High voltage plate.....	31
• High voltage plate > Ratings for fuses and safety switches	34
• High voltage plate > INRUSH print	36
• IPS module.....	36
• EPDAF/R print.....	38
• TRIAX print.....	40
• NG_LDEC	43
• NG_LDEC > OBMM110 print	52
• NG_LDEC > RBMM110 print	56
• NG_LDEC > CBMM110 print	60
• NG_LDEC > EBMM1020 print.....	62
• QM: main switch.....	64

• FL: mains filter (only with CE machines).....	64
• LR: coil	65
• T-ILLUM: transformer for fabric lighting	65
Main control box (CB) > Low-voltage section	66
• General	66
• Components	66
• MCB print	66
• DDMP print.....	71
• MIDB/R print.....	72
• MIDSB print.....	72
• IBR print	73
• UEB_X and UEB_Y	76
• Board rack.....	77
• Board rack > SBPO-print.....	77
• Board Rack > ATB_X, ATB_Y and ATB_Z print	78
TRIAx control box (CB-TRIAx)	81
• General	81
• Location.....	81
• T_TRIAX print.....	81
PICAN bus.....	86
• Function and description	86
• Signals and cable	87
• EPDAF print	87
• CJB print.....	89
• CAN and PICAN devices.....	89
• PICAN devices > Prewinders	90
• PICAN devices > PFL.....	92
• PICAN devices > Thermal cutter (eCTTH)	95
• PICAN devices > Jacquardette	96
Properties of the electrical components located outside the control boxes	97
• Display characteristics.....	97
• Properties of the proximity switches and other detection elements	97
• Properties of the stepping motors.....	98
• Properties of the servomotors	99
• Properties of the valves	99
• Properties of the safety brake.....	99
• Properties of the resolver	99
Diagrams	101
Standard diagrams.....	103
• CE machines without series transformer per machine	103
• CE machines with series transformer per machine	103
• Non CE machines without series transformer per machine.....	104
• Non CE machines with series transformer per machine	104
Diagrams for options	105
• Batching motion.....	105
• Outside cam motion with leveling	106
• Hot wire cutter (eCTTH) > Without centre hot wire cutters	106
• Hot wire cutter (eCTTH) > With 1 centre hot wire cutter	107
• Thermal cutter (eCTTH) > With 2 centre thermal cutters	107

• Fabric lighting above the fabric.....	108
• Fabric lighting underneath the fabric	108
• Secondary aspirator (MASP-2)	109
Power supply diagrams > General	110
• General > Power supplies – Overview	110
• General > Power Supply Control and Power fail	110
Power supply diagrams > +26V DC	111
• +26VDCS	112
• +26VDCS on the MCB print	112
• +26VDC1	113
• +26VDC2	114
• +26VDCPICAN	114
• +26VDC (in the case of ATB-prints configuration).....	115
• +26VDC (in the case of configuration without ATB-prints)	115
• +26VDC on the EB print	116
• +26VDC on the IBR print.....	116
• +26VDC on the UEB-X print (without ERGO).....	117
• +26VDC on the UEB-X print (with ERGO).....	117
• +26VDC on the UEB-Y print.....	118
• +26VDC on the ATB_X print	118
• +26VDC on the ATB_Y print	119
• +26VDC on the ATB_Z print.....	119
Power supply diagrams > 290V	120
Power supply diagrams > 48V AC	121
Power supply diagrams > 41V DC	121
Power supply diagrams > +12VDC	122
Power supply diagrams > -12VDC	122
Power supply diagrams > +5VDC	123
Connecting the machine terminal and the diagnostic plug	124
Emergency stop diagrams	125
• Emergency stops in case of dobby or outside cam motion.....	125
• Emergency stops in case of dobby or outside cam motion with double beam.....	126
• Emergency stops in case of jacquard.....	126
• Emergency stops in case of jacquard with double beam.....	127
Control button diagrams	128
• Control buttons > Detections	128
• Control buttons > Protection logic (second contact).....	129
Signals diagrams	130
• Signal lights.....	130
• Warning lights	130
• Warning lights only with double beam	131
PICAN diagram	132
• PICAN diagram > with jacquard and without Picanol-PFL.....	133
• PICAN diagram > with jacquard and with Picanol-PFL.....	133
• PICAN diagram > dobby or outside cam motion without Picanol-PFL.....	133
• PICAN diagram > dobby or outside cam motion and Picanol-PFL	134
• PICAN diagram > jacquardette CX180	134

• PICAN diagram > DWC	135
• PICAN diagram > CAN doobby	135
• PICAN diagram > jacquard	136
Internal connections	137
Cables without number	137
• Cables without number > part 1	138
• Cables without number > Cables on the QSC back rest	138
Cables sorted according to number	143
• #52: Resolver	145
• #55: Oil pressure detection	145
• #57A: Detection levelling outside cam motion	145
• #60: Parallel communication	145
• #66: Signal light on the control box	145
• #67A: High voltage in control box	146
• #67B: High voltage in control box	146
• #101: Keyboard and display	146
• #102: Keyboard and display	146
• #105A: Signal light on the machine	146
• #105B: Signal light on the machine	147
• #108: POWER ON push buttons on the control box	147
• #116: Water temperature detection	147
• #117: Oil temperature detection	147
• #171: RHS airtucker valves	147
• #172: RHS airtucker cutter	147
• #174: Proximity switch in pickfinding position	148
• #175: Proximity switch in weaving position	148
• #188 #189 #190: ELO 1 motor	148
• #191 #192 #193: ETU motor	149
• #194 #195 #196: ELO 2 motor	150
• #203: Back rest QSC standard	150
• #204: Back rest QSC load cell 1	151
• #205: Back rest QSC load cell 2	151
• #207: Thermal detection in aspirator motor	151
• #209 #210 #211: ELO-FA motor	151
• #215: Dobby	152
• #220: ELSY LHS	153
• #221: ELSY RHS	153
• #222: Waste cutter LHS	153
• #272: Thermal detection in oil pump	153
• #285A: Fabric cutter 1	153
• #285B: Fabric cutter 1	153
• #286A: Fabric cutter 2	154
• #286B: Fabric cutter 2	154
• #287A: Fabric cutter 3	154
• #287B: Fabric cutter 3	154
• #288A: Fabric cutter 4	154
• #288B: Fabric cutter 4	154
• #320: Diagnostic plug on the control box	155
• #362: ELSY GLASS LENO	155

• #378: Hydraulic control.....	155
• #417: Quickstep motor 1 up to and including 4.....	155
• #418: Quickstep motor 5 up to and including 8.....	156
• #419: Quickstep motor 9 to 12	156
• #420: Filling detector 1	157
• #421: Filling Detector 2	157
• #424 #425 #426: OPTILENO motor	157
• #429: DD to the NG_LDEC print	158
• #430: Reference proximity switch	158
• #431: Waste cutter RHS.....	158
• #432: Rotating filling cutter	158
• #497: IRO rewinder cable with PIKAN communication	159
• #498: LGL rewinder cable with PIKAN communication	159
• #504 #505 #506: ELO2 motor	160
• #510: Centre ELSY 1	160
• #511: Centre ELSY 2	160
• #512: Centre ELSY 3	160
• #513: Gripper clamp cleaning valve	161
• #516: LHS airtucker cutter.....	161
• #518: Warp stop motion LHS and RHS detection	161
• #519: Safety switch fancy NQSC	161
• #523: Leno Detection LHS NQSC	161
• #524: Leno Detection RHS NQSC	162
• #527 #528 #529: ELOFA motor.....	162
• #537: LHS airtucker valves	162
• #538: Airtucker valves centre 1	163
• #539: Airtucker cutter centre 1	163
• #540: Airtucker valves centre 2	163
• #541: Airtucker cutter centre 2	163
• #542: Airtucker valves centre 3	163
• #543: Airtucker cutter centre 3	163
• #562A: PKE 12C	164
• #562B: PKE *C.....	166
• #566: Power supply for the MCB print inside the control box	167
• #567: Power supply for the MCB print inside the control box	167
• #568: Power supply (5V; 12V; 26V) for the IBR print in the control box.....	167
• #569: Power supply for the SBPO print inside the control box	168
• #570: ESAC & POWER ON signal on the control box.....	168
• #571: Emergency stop and POWER ON push button	168
• #573: Power supply (12VDC; 48VAC) for the EPDAF print in the control box.....	168
• #574: PIKAN communication LHS	169
• #574AA: PIKAN LHS.....	169
• #574AB: PIKAN LHS.....	169
• #574A1: PIKAN FD 8C.....	170
• #574A1: PIKAN FD 12C.....	170
• #574B: PIKAN communication LHS.....	170
• #574C: PIKAN Intermediate cable	170
• #575: PIKAN communication RHS.....	171
• #576: PIKAN communication Medium Speed for the EPDAF print	171
• #577: Pulse train and eventline for the EPDAF print	171
• #579: Thermal cutter module communication	171
• #582: Connection between the EPDAF and the MCB print in the control box	171

• #583: EPDAF to the DDMP print in the control box	171
• #584: Safety brake	172
• #585: Power supply (290VDC; 26VDC) for the (PICAN) prewinders.....	172
• #586: Power supply (290VDC) for the TRIAX print in the control box.....	172
• #588: Power supply (26VDC, 41VDC) for the IBR print in the control box.....	173
• #589: Power supply (26VDC) for the UEB_X print inside the control box.....	173
• #590: Power supply (26VDC) for the UEB_Y print inside the control box.....	173
• #591: Power supply (26VDC) for the ATB_X print inside the control box.....	173
• #592: Power supply (26VDC) for the ATB_Y print inside the control box.....	174
• #593: Power supply (26VDC) for the ATB_Z print inside the control box.....	174
• #594 #595 #596: Power supply for the TRIAX print inside the TRIAX box.....	174
• #597: 290V power supply for the EPDAF print inside the control box.....	175
• #598: Electronic RHS gripper opener (ERGO2)	175
• #605A: Emergency Stop Button and Warning Light	175
• #608: IPS Detection	176
• #633: Power Supply (26VDC) to EB in the main Control Box.....	176
• #653: Left-Hand side Emergency Stop Button and Warning Light.....	176
• #655: Gripper flushing valve.....	176
• #663A: Aspirator motor 2 and thermal detection	177
• #663B: Aspirator motor 2 and thermal detection	177
• #665: CAN communication between the MCB and the TRIAX in the control box.....	178
• #672: NG_LDEC to SUMO.....	178
• #686: Power supply (26VDC) to EB board in the main control box (small rack).....	178
• #692: Detection lubrication filter	179
• #704: Electrical valve for water cooling	179
• #706: Detector Warp Tension (DTWT1).....	179
• #707: Detector Warp Tension (DTWT2).....	179
• #708: Detector Warp Tension (DTWT3).....	179
• #710: Let-off motion and pull through push-buttons on control box.....	179
• #711: Push-buttons fancy beam (PBFA)	180
• #716: PICAN RHS.....	180
• #719: Wasteless 2C	180
• #733 #734 #735 ETU motor in case of extra TRIAX and IPS	180
• #736 #737: IPS-BOX.....	181
• #741: LED-light ILUP RHS	181
• #741B 742B: LED ILLUM RHS.....	181
• #742: LED-light ILLow RHS	181
• #743: LED-light ILUP LHS.....	181
• #744: Warp detection compensator unit.....	182
• #745: Warp detector draw in	182
• #746: Detection pressure roller LHS	182
• #747: Detection pressure roller RHS	182
• #748: Jacquardette.....	182
• #749: Emergency stop button on the jacquard or jacquardette	182
• #750: Power supply (5V, 12V & 26V) & Drive Disable signal to Triax in the main control box.....	183
• #751: LStop Pwr (WYD+sup)	183
• #752: LStop Takt (Z-pulse).....	183
• #753: LED-light ILLow LHS.....	183
• #754: Power Supply (26VDC) to relay ILUP	183
• #755: Power Supply (26VDC) to relay ILLow.....	183
• #756: PICAN EPDAF to OXI-PIC(A)	183
• #757: PICAN OXI-PIC	184
	184

• #759: High Performance Waste Cutter Left-Hand side (HPWC LHS)	185
• #760: High Performance Waste Cutter Right-Hand side (HPWC RHS)	185
• #761: High Performance Cutter Middle 1 (HPWC MID1)	186
• #762: High Performance Cutter Middle 2 (HPWC MID2)	186
• #763: High Performance Cutter Middle 3 (HPWC MID3)	186
• #764: High Performance Cutter Middle 4 (HPWC MID4)	187
• #765: SmartCut	188
• #766 #767 #768: ELO Fancy Motor	188
• #772: Lstop S1 (TX1/shed)	188
• #773: Lstop E1 (RX1/shed)	189
• #774: Lstop S2 (TX2/Low)	189
• #775: Lstop E2 (RX2/low)	189
• #779: LED-light ILUPFA LHS	189
• #780: LED-light ILUPFA RHS	189
• #788: Transmitter Hand Safety OMRON	189
• #789: Receiver Hand Safety OMRON	190
• #790: PS_48VDC Supply	190
• #791: OXI 48VDC Supply	190
• #792: Pican eCTTH(OXI)	190
• #793A: PUSH BUTTONS machine	190
• #793B: Push buttons Machine	191
• #793B: RHS push buttons to LHS push buttons	191
• #794: ELECTRONIC RIGHT-HAND SIDE GRIPPER OPENER 1 motor (ERGO)	191
• #795: SLEY BLOWING	191

ections	193
• ATB_X print	193
• ATB_Y print	194
• ATB_Z print	195
• DDMP print	196
• DWC Control	196
• EB (Extension Board)	197
• EPDAF print	198
• IBR print	205
• IPS module	208
• NG_LDEC	209
• MCB print	211
• MIDB/R print	218
• MIDSB print	218
• SBPO print	219
• TRIAX print	220
• T_TRIAX print	223
• UEB-X print	227
• UEB-Y print	229

Pictographs, abbreviations and safety devices

Contents

This section contains the following subjects:

• Pictographs used	9
• Abbreviations	9
• Safety Devices	9

Pictographs used

Pictograph	Name	Description
	Note	Provides additional information concerning important topics in the text.
	Warning	Disregarding this warning may result in the loss of electronic data or in mechanical damage.
	Danger	Disregarding this warning may lead to material damage but also and in particular to physical injury to persons.
	CE version	Only applies to the CE version.
	Liability	Provides information about the legal liability.

Abbreviations

Introduction

In this section, the abbreviations for the colors of the cable wires are determined. Other abbreviations are explained in the 'Abbreviations' section in the Picanol manual.

Single-colored wires

The following table gives an overview of the abbreviations

Abbreviation	In full	Description
BK	Black	black
BL	BLue	blue
BR	BRown	brown
GR	GReen	green
GY	GreY	grey
OR	ORange	orange
PK	PinK	pink
RD	ReD	red
VT	VioleT	violet
WH	WHite	white
YW	YelloW	yellow

Earth wire

The following table gives an overview of the abbreviations.

Abbreviation	In full	Description
YWGR	YelloWGReen	yellow/green earth wire

Two-colored wires (except YWGR)

The following table contains some examples.

Abbreviation	In full	Description
GY/PK	GreY/PinK	grey wire with pink rings
WH/RD	WHite/ReD	white wire with red rings

Safety Devices

Operating instructions



Before performing any action, carefully read the Operating Instructions!

Emergency stop



The control box remains live when the emergency stop button is pressed.

- This is a red button on a yellow background.
- If one emergency stop is not unlocked, the control buttons will not work.
=> unlock = 1/8 turn clockwise.

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Safety light grid



The safety light grid does NOT function when the machine is operating at FULL speed.

- Is located over the pressure roll.
- The safety light grid prevents movement of the sley during start-up, slow motion and pick finding. The machine motion can be started again following the normal procedure.

STAND-BY

- Must be locked in this position for all repair activities and lubrication work.
- Position for setting the machine.
- All machine movements are blocked, but the electronic components will remain live.

POWER ON

- Power circuits are turned on.
- All motors, except for the rewinder motor and the oil pump, remain switched off.
- The main motor will only start after:
 - The start button has been pressed;
 - An automatic movement has been performed;
- Machine start-up is indicated by flashing warning lights.
- In case of emergency, all machine actions can be interrupted by means of an emergency stop.

Warning lamps

If the warning lamps are flashing, do not touch the weaving machine.

Main switch

- Is located on the control box => high-voltage section door.
- Must be padlocked in the -0- position when major work is performed.

Work to be performed

- The control boxes may only be opened and connected by qualified and authorized members of staff. This means that they must have a basic knowledge of the fundamental rules of electricity and the relevant legislation, and that they must have experience with electric wiring, electric components and with the symbols on electric diagrams.
- If, for work to be performed, it is required that the earth connections are disconnected, then these must be reconnected in the original way before the control system is switched on again.
- When a control box is open, EMC (electromagnetic compatibility) is not guaranteed.
- While work is going on with a control box open and the main switch on, no appliances producing strong electromagnetic fields – such as mobile phones or walkie-talkies – may be brought near the control box.
- The transparent plastic covers may never be removed before switching off the power first.

Earth connection

- The weaving mill must have an effective earth connection. This must have been checked and verified by the services responsible.
- After the control box and the rewinder standard have been placed, the earth wire must first be connected to the machine. For more information on the earthing, see the 'Installation' section.
- The machine must be connected to the mains earth connection in the prescribed manner.
- The earth connections interconnecting machine components may not be interrupted. If during work on the machine they are disconnected, they must be reconnected in the original way before the control system is switched on again.

Stepping motors

- The machine is equipped with several stepping motors: at the filling cutters, the waste cutters, the selvedge motions...
- These motors may never be opened, as some loss of torque may result from opening and reclosing them.
- In the event of a fault or failure, these motors should be sent to Picanol.

General

Contents

This section contains the following subjects:	13
Connection conditions	13
• Power supply cable specifications	15
• Supply cable protections	17
Control system and machine drive	17
• Principle	17
• Security	18
• Power supply - motor switches	18
• Control system of the main motor control box	18
• Main motor	18
• Brake	19
• Cooling system	19
• Pickfinding	20
• Weaving (driving the machine)	20
• PXWP and PXFP	21
• Let-off	25
Overview of the weaving machine	25
• Overview of weaving machine 1	25
• Overview of weaving machine 2	26
• Overview of weaving machine 3	26
• Overview of weaving machine 4	27
Main control box (CB) > Parts on the outside of the main control box	27
• Overview of the outside of the main control box	27
• Side 1	28
• Side 2	28
• Side 3	29
• Side 4	30
Main control box (CB) > High-voltage section	30
• General	31
• Components	31
• High voltage plate	34
• High voltage plate > Ratings for fuses and safety switches	36
• High voltage plate > INRUSH print	36
• IPS module	38
• EPDAF/R print	40
• TRIAX print	43
• NG_LDEC	52
• NG_LDEC > OBMM110 print	56
• NG_LDEC > RBMM110 print	60
• NG_LDEC > CBMM110 print	62
• NG_LDEC > EBMM1020 print	64
• QM: main switch	64
• FL: mains filter (only with CE machines)	65
• LR: coil	65
• T-ILLUM: transformer for fabric lighting	66
Main control box (CB) > Low-voltage section	66
• General	66

• Components	66
• MCB print	71
• DDMP print	72
• MIDB/R print	72
• MIDSB print	73
• IBR print	76
• UEB_X and UEB_Y	77
• Board rack	77
• Board rack > SBPO-print	78
• Board Rack > ATB_X, ATB_Y and ATB_Z print	81
TRIAx control box (CB-TRIAx)	81
• General	81
• Location	81
• T_TRIAX print	81
PICAN bus	86
• Function and description	86
• Signals and cable	87
• EPDAF print	87
• CJB print	89
• CAN and PICAN devices	89
• PICAN devices > Prewinders	90
• PICAN devices > PFL	92
• PICAN devices > Thermal cutter (eCTTH)	95
• PICAN devices > Jacquardette	96
Properties of the electrical components located outside the control boxes	97
• Display characteristics	97
• Properties of the proximity switches and other detection elements	97
• Properties of the stepping motors	98
• Properties of the servomotors	99
• Properties of the valves	99
• Properties of the safety brake	99
• Properties of the resolver	99

Connection conditions

Contents

- Power supply cable specifications [13]
- Supply cable protections [15]

Power supply cable specifications

Contents

- Load demands [13]
- Rated power [13]
- General conditions [14]
- Electrical requirements of the weaving room [14]
- Cable section [14]

Load demands

Power consumption is the energy consumption during normal machine operation.
The power consumption depends on:

- the machine configuration,
- the machine speed,
- the type of fabric and the machine settings.



The required load must be calculated for each machine separately.

Rated power

The rated power depends on external factors during a cold start or in specific working environments.



The mains must always be able to supply enough power in similar extreme situations.

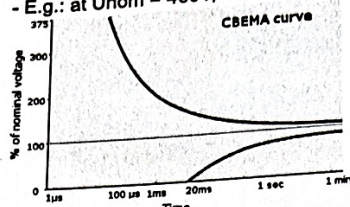


The rated power is also taken into account when calculating the mains cable conductor size.

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General conditions

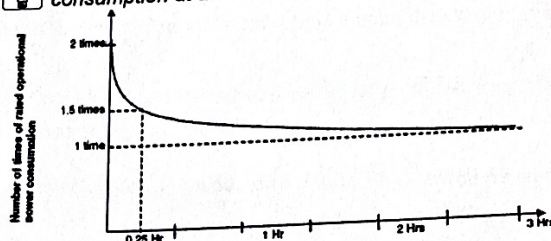
- **Stability of the power supply**
 - Supply voltage fluctuations must be between -10% and +10% of the nominal operating voltage (EN50160).
 - E.g.: at $U_{nom} = 460V$, the supply voltage must never drop below 414V or be higher than 506V.



- **Power consumption at cold start**
At a cold start (e.g. after a weekend stop) power consumption is twice as high as compared to the consumption at typical operating temperature conditions. It takes about 3 hours for the power consumption to drop to normal.



The electrical wiring and the transformer (if present) must be able to support this high power consumption at a cold start.



- **Conditions at machine start-up**
At machine start-up, the power consumption is approximately 80KVA (for 0.2sec). During this short time lap, the voltage at the machine should not drop below 85% of the nominal operating voltage.

Electrical requirements
of the weaving room

The weaving machines need a nominal voltage between 380V and 460V.

If the weaving room does not dispose of a power supply between these values, a transformer must be used.

Cable section

The tables below show the minimum value of the cable section for a specific supply voltage and the maximum cable length from the power source to the machine. These tables take the above mentioned requirements into account.

These tables apply if:

- every machine has a separate supply cable to the power source.
- The supply voltage equals the nominal level.



If the supply voltage drops below the nominal voltage, the cross section of the connection cable must be increased to compensate for that loss in voltage.

For the calculation of the cable cross section, the following resistances are taken into account (copper wires):

Cable section in mm ²	Resistance per meter in Ω
50	0,000387
35	0,000524
25	0,000727
16	0,00115
10	0,00183
6	0,00308
4	0,00461

The table below is used for **stable supply voltages** (which meet the European standard EN50160), or corresponding standard that apply to your country.

cable section (mm²)		Supply voltage (V)														
		200	220	260	280	346	380	400	415	440	460	480	500	550	575	600
Cable length power source - machine (meter)	10	6	4	4	4	4	4	4	4	4	4	4	4	4	4	4
	20	10	10	6	6	4	4	4	4	4	4	4	4	4	4	4
	30	16	16	10	10	6	4	4	4	4	4	4	4	4	4	4
	40	25	16	16	10	6	6	6	4	4	4	4	4	4	4	4
	50	35	25	16	16	10	6	6	6	6	6	4	4	4	4	4
	60	50	35	25	16	10	10	10	6	6	6	6	6	4	4	4
	70	70	50	35	25	16	10	10	10	10	6	6	6	6	4	4
	80	100	70	50	35	25	16	10	10	10	10	6	6	6	6	4
	90	150	100	70	50	35	25	16	10	10	10	10	10	6	6	6
	100	200	150	100	70	50	35	25	16	10	10	10	10	10	6	6

The table below is used for power voltages where **voltage fluctuations** may be expected (the fluctuations may not exceed 10%).

cable section (mm²)		Supply voltage (V)														
		200	220	260	280	346	380	400	415	440	460	480	500	550	575	600
Cable length power source - machine (meter)	10	10	10	6	6	4	4	4	4	4	4	4	4	4	4	4
	20	25	25	16	16	10	6	6	6	6	4	4	4	4	4	4
	30	45	25	25	16	10	10	10	10	10	6	6	6	4	4	4
	40	50	35	25	25	16	16	10	10	10	10	10	10	6	6	6
	50	70	50	35	35	25	16	16	16	16	10	10	10	10	10	10
	60	70	70	50	35	25	25	16	16	16	16	16	10	10	10	10
	70	95	70	50	50	25	25	25	25	16	16	16	16	10	10	10
	80	95	70	50	50	25	25	25	25	25	16	16	16	16	10	10
	90	95	95	70	50	25	25	25	25	25	25	16	16	16	16	10
	100	120	95	70	50	25	25	25	25	25	25	25	16	16	16	16



When using one cable to connect the main distribution point with multiple machines and another distribution point is available close to the machines,

- the copper of the cables can be used in a more efficient way as the cable is rated with a higher voltage. As a result, the difference between nominal power and peak power at start-up of one of those machines becomes smaller.
- In most cases, there is no influence on the nominal voltage during start-up of a machine. The nominal voltage can drop 15% during startup.

Supply cable protections

Contents

- Overload protection [15]
- Icu value of the control box [15]
- Leak current protection [16]

Overload protection

The overload protection upstream the power cable of the weaving machine protects:

- the power supply cable,
- the main switch in the control box.



The thermal protection can be done with fuses of the type aM (according to the standard IEC 60269-1).

The table below shows the fuses to use:

Supply voltage	Value of the type aM fuse
200V to 346V	40
380V to 480V	25
500V to 600V	25

Icu value of the control box

The maximum allowed short circuit current (Icu) supplied to the main switch, is limited, according to the standard IEC 60947-2). The power distribution must be able to cope with that. The maximum allowed short circuit current (Icu) can be found in the table below.

Supply voltage	Icu (A)
200V to 346V	10.000
380V to 480V	15.000
500V to 600V	4.000

**Leak current
protection**

The machine control has no leak current protection.
If the local laws demand a leak current protection, it must be provided in the power distribution of the weaving mill.

The leak current protection should have the following properties:

- time delayed leak current protection
- should not be lower than 300mA.
- one leak current protection per machine.

Control system and machine drive

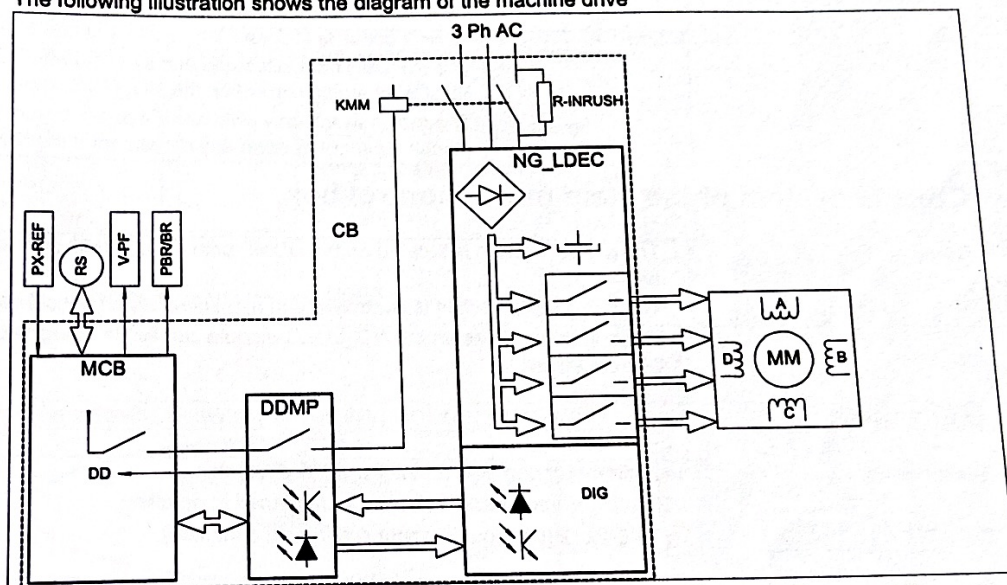
Contents

- Principle [17]
- Security [17]
- Power supply - motor switches [18]
- Control system of the main motor control box [18]
- Main motor [18]
- Brake [18]
- Cooling system [19]
- Pickfinding [19]
- Weaving (driving the machine) [20]
- PXWP and PXPFP [20]
- Let-off [21]
- Take-up motion [24]

Principle

Diagram

The following illustration shows the diagram of the machine drive



Security

Protection

The safety system combines software and hardware protection.

The hardware protection consists of:

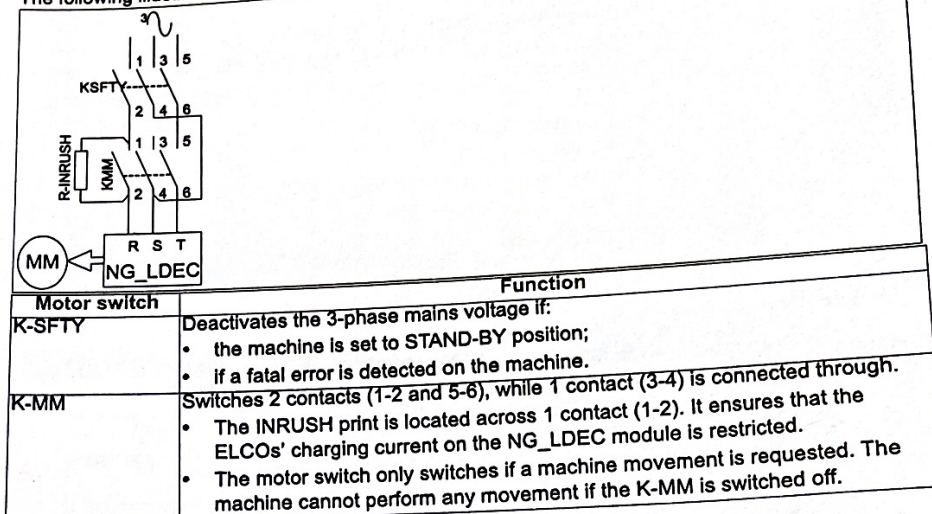
- the second contact (PB*-2) (where * = SMF, SMR, PFF or PFR) of the buttons controlling machine movements. See Control buttons > Protection logic (second contact) [129];
- the activation of the STAND-BY position;
- the safety light grid;
- the relay control unit on the MCB print;
- the emergency stops.

The safety systems on the MCB print and the DDMP print are part of the total machine safety system.

Power supply - motor switches

Diagram

The following illustration shows the diagram of the power supply



Control system of the main motor control box

General

- The control system is located on the DDMP print. The DDMP print is plugged in on the MCB print.
- The NG_LDEC module is controlled and monitored via an optical connection.
- The control modules on the NG_LDEC module can be de-energized by means of the Drive Disable signal.

Main motor

Stator coils

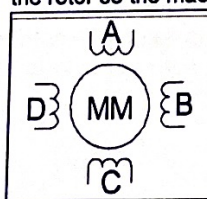
The main motor contains 8 stator coils (4 double):

- 2 by 2 coils in opposite positions are placed in parallel.
- $R = 0.22 \Omega$ (per double stator coil in cold condition).

Rotor shaft

The motor shaft moves radially and is driven by 4 double stator coils.

- When the machine is active, the stator coils are driven in such a manner that the rotor rotates.
- In case of braking, the stator coils are driven in such a manner that they exert a braking effect on the rotor so the machine stops.



Brake

Machine running

The running machine is braked by controlling the main motor stator coils in a way resulting in a braking effect on the rotor shaft.

Machine stopped

When the machine is stopped, it is held motionless by a safety brake.

- When the brake is actuated, the machine is loose.
- At power OFF, the machine is held.

LET'S GROW TOGETHER Cooling system

OptiMax-i

Oil

An oil circuit of the circulating lubrication cools the stator of the motor.

Heatsink

The following components of the NG_LDEC module are fitted against the heatsink:

- the power control system control modules composed of 2 OBMM prints;
- the RBMM print (3-phase rectifier);
- the dump resistor.

Water cooling

The electric valve "A" is controlled by the main control box, based on the information gathered from the oil temperature sensor.

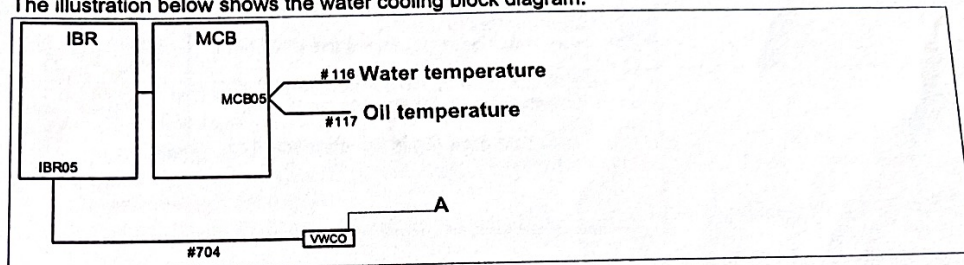
As long as the oil temperature does not reach 37°C the water cooling valve remains closed in order to let the machine achieve full operation temperature, thus reducing power consumption.

Once 37°C is reached the main control box opens the water cooling valve "A" and leaves it open until the oil temperature sinks to 35°C. At 35°C the valve is closed again until the oil temperature reaches 37°C again.

During the stop of the machine the operation of the water cooling valve remains as described above. This way the machine oil is not cooled below 35°C in order to hold the operational temperature as long as possible (energy saving).

Block diagram water cooling

The illustration below shows the water cooling block diagram:



Pickfinding

Process

The following illustration shows the pickfinding process;

